provide data to other systems.

Sensors and devices can connect indirectly through the cloud, where data is centrally-managed, or send data directly to other devices to locally collect, store and analyse the data, and then share selected findings or information with the cloud. Edge devices for data management help secure the most valuable data and reduce bandwidth cost. These also provide great performance, ownership over data and lower maintenance cost.

Edge devices run a Web-based dashboard that end-users can access to monitor the flow of data, so they can decide how various systems in demonstration and devices are running, and get notified by alarms. A large amount of data can be represented in the form of a graph for any desired range of time, and each point on the graph represents a record that can be found by searching the database, which stores a large quantity of data.

AWS IoT device management. This is critical across industrial, consumer and commercial applications such as an industrial and connected home. It enables customers to manage large and diverse device fleets such as operational technology systems, cameras, machines, vehicles, appliances and more.

It enables monitoring of usage and performance metrics for devices across industrial sectors such as manufacturing, oil and gas, and mining. It allows monitoring metadata and policy changes with service alerts to inform about any adjustments to the devices' configuration. It also allows detection of any unusual behaviour across a device and to take mitigating actions.

It provides secure onboarding, organising, monitoring, troubleshooting and sending of firmware updates over-the-air (OTA). It enables adding device attributes like device name, type and manufacturing year, certificates and access policies to the IoT Registry. Then, it assigns them to devices, and makes connected devices ready for service quickly. This helps to quickly search and find any IoT device across the entire device fleet in near real time.

It is easy to find devices based on a combination of attributes like device ID, device state and type, so that action can be taken, in addition to troubleshooting. Actions such as reboots, factory resets and security patches can also be remotely executed.

Data management challenges

With time, the number of IoT devices will increase, thus increasing the challenges for real-time processing and analysis to reduce time for storage. Space has to be optimised for metadata like user IDs and passwords to ensure enough space for new information.

Functions such as adaptive maintenance, predictive repair, security monitoring and process optimisation rely on real-time data. Selecting the right tools is a challenge because integration between different sensors should be proven and compatibilities confirmed. When there is no connection, devices must still gain insights, make decisions and prepare for data distribution.